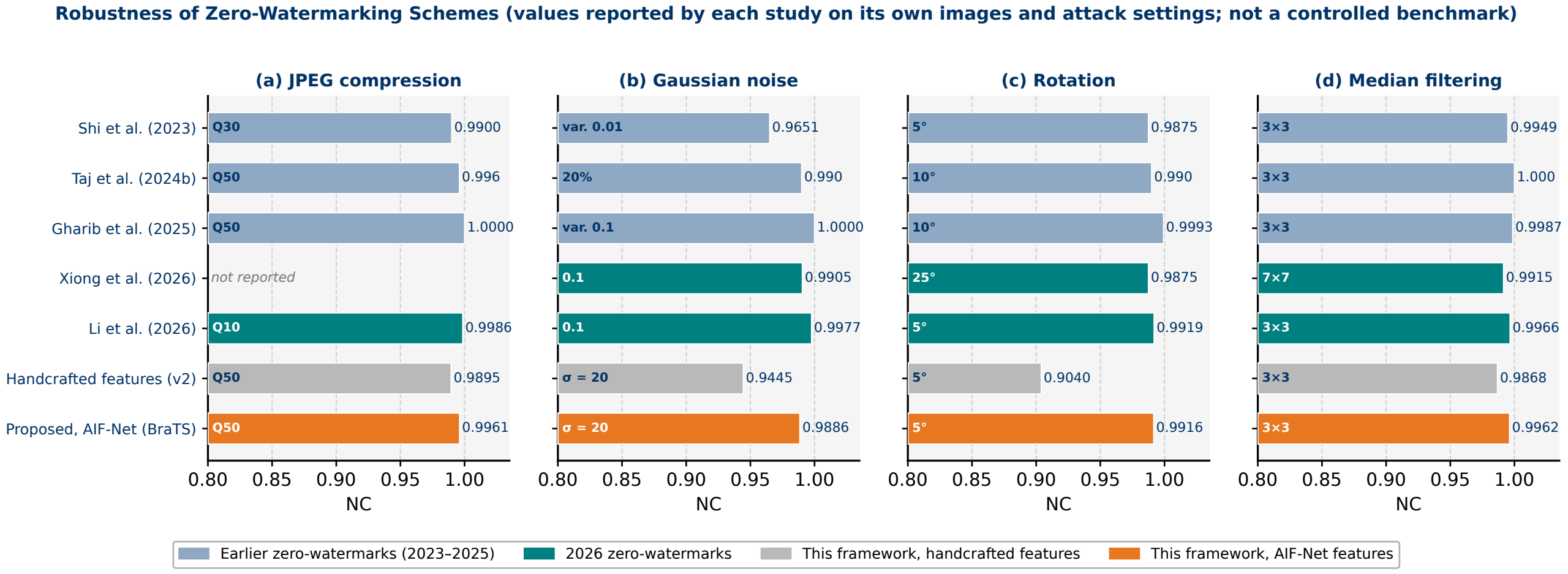
Robustness of Zero-Watermarking Schemes (values reported by each study on its own images and attack settings; not a controlled benchmark)
(a) JPEG compression
(b) Gaussian noise
(c) Rotation
(d) Median filtering
Shi et al. (2023)
Taj et al. (2024b)
Gharib et al. (2025)
Xiong et al. (2026)
Li et al. (2026)
Handcrafted features (v2)
Proposed, AIF-Net (BraTS)
Q30
0.9900
Q50
0.996
Q50
1.0000
not reported
Q10
0.9986
Q50
0.9895
Q50
0.9961
var. 0.01
0.9651
20%
0.990
var. 0.1
1.0000
0.1
0.9905
0.1
0.9977
σ = 20
0.9445
σ = 20
0.9886
5°
0.9875
10°
0.990
10°
0.9993
25°
0.9875
5°
0.9919
5°
0.9040
5°
0.9916
3×3
0.9949
3×3
1.000
3×3
0.9987
7×7
0.9915
3×3
0.9966
3×3
0.9868
3×3
0.9962
0.80
0.85
0.90
0.95
1.00
NC
Earlier zero-watermarks (2023–2025)
2026 zero-watermarks
This framework, handcrafted features
This framework, AIF-Net features